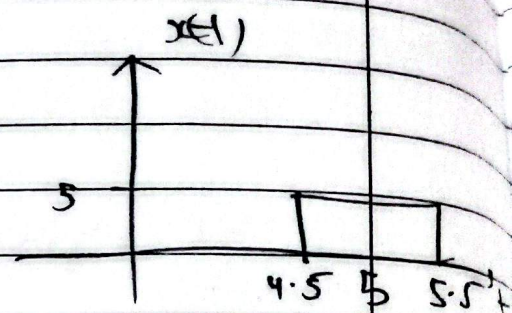


Quiz #05 SNL - Spring 2026.
FT

$$x(t) = \text{rect}(t - 5)$$

$$\text{Find } X(\omega) = ?$$



$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{4.5}^{5.5} (5) e^{-j\omega t} dt$$

$$= \frac{-5j\omega}{j\omega} e^{-j\omega t} \Big|_{4.5}^{5.5}$$

$$= \frac{-5}{j\omega} (e^{-j5.5\omega} - e^{-j4.5\omega})$$

$$= \frac{-5}{j\omega} e^{-j5\omega} (e^{-j0.5\omega} - e^{j0.5\omega})$$

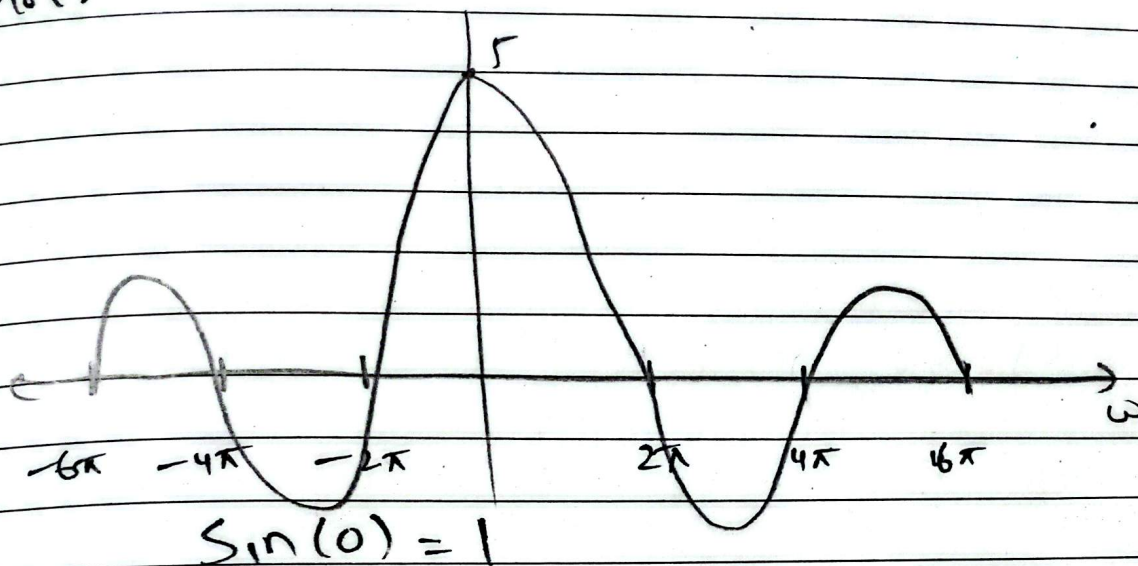
$$= \frac{5}{j\omega} e^{-j5\omega} (e^{j\omega/2} - e^{-j\omega/2})$$

$$= \frac{5}{j\omega} e^{-j5\omega} (2j \sin(\omega/2))$$

$$= 5 e^{-j5\omega} \frac{\sin(\omega/2)}{(\omega/2)}$$

$$= 5 e^{-j5\omega} \text{sinc}(\omega/2) \text{ proved}$$

plots



$$\sin(x) = 0 \quad @ \quad x = \pm n\pi$$

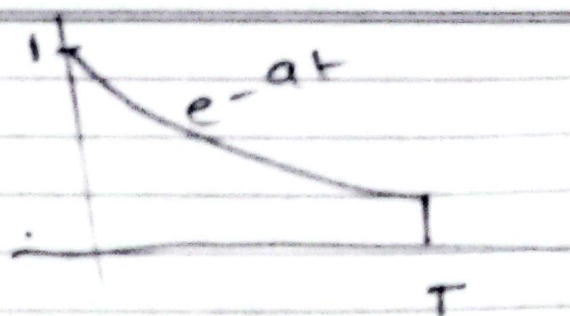
in given case.

$$\omega/2\pi = \pm n\pi$$

$$\omega = \pm 2n\pi$$

$$\therefore \text{Zero crossing @ } \omega = \begin{matrix} \pm 2\pi, & 4\pi, & 6\pi \\ n=1 & n=2 & n=3 \end{matrix}$$

2.



$$x(t) = e^{-at} [u(t) - u(t-T)]$$

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \text{Using property.}$$

$$x(t) = e^{-at} u(t) - e^{-at} u(t-T)$$

Apply linearity property.

$$X(\omega) = \mathcal{L} \{ e^{-at} u(t) \} - \mathcal{L} \{ e^{-at} u(t-T) \}$$

→ F.T pair e^{-at}

$$e^{-at} u(t) \leftrightarrow \frac{1}{a + j\omega}$$

→ F.T time shifting property.

$$x(t-T) \leftrightarrow X(\omega) e^{-j\omega T}$$

$$= \mathcal{L} \{ e^{-at} u(t) \} - \mathcal{L} \{ e^{-at} u(t-T) \}$$

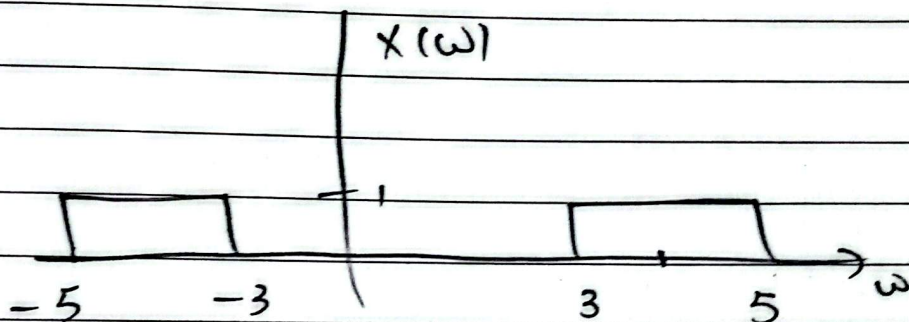
$$= \mathcal{L} \{ e^{-at} u(t) \} - \mathcal{L} \{ e^{-at} \cdot e^{-j\omega T} u(t-T) \}$$

$$\therefore X(\omega) = \frac{1}{a+j\omega} + e^{-aT} \frac{1}{a+j\omega} e^{-j\omega T}$$

$$= \frac{1}{a+j\omega} \left[1 - e^{-(a+j\omega)T} \right]$$

proved

3.



$$X(\omega) = \text{rect}\left(\frac{\omega-4}{2}\right) + \text{rect}\left(\frac{\omega+4}{2}\right)$$

F-T Pair.

$$\text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

in given $X(\omega)$ $\tau = 2$, shifting factor = 4.

$$x(t) = \mathcal{L}^{-1}\left\{\text{rect}\left(\frac{\omega-4}{2}\right)\right\} + \mathcal{L}^{-1}\left\{\text{rect}\left(\frac{\omega+4}{2}\right)\right\}$$

$$= 2\text{sinc}\left(\frac{t \cdot 2}{2}\right) e^{-j4t} + 2\text{sinc}\left(\frac{t \cdot 2}{2}\right) e^{j4t}$$

$$= 2\text{sinc}(t) (e^{-j4t} + e^{j4t})$$